



An Internship Report
Of
Data Science Master Virtual Internship

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

By

Roll Number of Student: 2410030229

Name of Student: VINEET GUPTA

Semester: 5th Section: 3CSE2

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, VINEET GUPTA, do hereby declare that the internship report titled "Data Science Master Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: VINEET GUPTA

Roll No.: 2410030229

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE - EduSkills Virtual Internship Program team and EduSkills Academy, along with Siemens Industry Software, for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: VINEET GUPTA

Roll No.: 2410030229

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3. INTERNSHIP COMPLETION CERTIFICATE



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Certificate of Virtual Internship

This is to certify that

Vineet Gupta

IILM University, Greater Noida

has successfully completed the 8-weeks

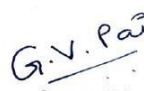
Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

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Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 44d7b3aef729cae3c5be

Student ID: STU6835624f24a3a1748329039



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

5

Internship Offer Letter



Ref No: C16Y2026M091441050

Internship Offer Letter

Student Name : Vineet Gupta
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU6835624f24a3a1748329039

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.


Bibek Ranjan
 Director, EduSkills

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE - EduSkills Virtual Internship Program, in the domain of "Data Science Master", supported by Siemens Industry Software. The internship was carried out from April 2026 to June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a structured, industry-oriented foundation in data science - covering data engineering, machine learning, and enterprise analytics platform administration using the RapidMiner AI Hub ecosystem. The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, and project documentation, culminating in a Final Assessment Test. On successful completion of all the requirements, a Virtual Internship Certificate (Grade: O - Outstanding) was awarded by EduSkills Academy, in association with Siemens Industry Software (India) Pvt. Ltd. and AICTE.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme "Nation Building Through Skills". EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE - EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal, Ministry of Education. For this internship track, EduSkills collaborated with Siemens Industry Software, which supported the Data Science Master curriculum built on the RapidMiner AI Hub platform (an Altair company), enabling engineering students to gain practical exposure to enterprise-grade data science and machine learning tools.

4.3 Problem Statement

While academic coursework introduces students to the theory of data science and machine learning, most engineering curricula provide limited hands-on exposure to complete, enterprise-grade data science platforms used in the industry. Concepts such as structured data engineering pipelines, advanced text processing, model deployment,

and platform administration are rarely practiced in a real tool environment. The problem addressed by this internship was to bridge this gap by providing structured, hands-on training on an industry data science platform (RapidMiner), progressing from foundational data access and transformation skills to advanced machine learning, model deployment, and enterprise platform administration.

4.4 Project Objectives

- To understand how to identify business problems suitable for AI and machine learning solutions.
- To learn data access, transformation, and multi-dataset handling using RapidMiner (Data Engineering track).
- To build foundational and advanced machine learning skills, including classification, regression, clustering, and association rules.
- To master advanced data preparation techniques such as loops, branches, regular expressions, and web APIs.
- To learn advanced machine learning techniques including cross-validation, parameter optimization, feature engineering, and time-series forecasting.
- To gain practical experience in deploying and monitoring machine learning models and web applications.
- To learn RapidMiner AI Hub platform administration, including installation, Radoop, real-time scoring, and marketplace management.

4.5 Scope of the Project

The scope of this internship spans the complete data science workflow taught through the RapidMiner platform - starting from identifying AI/ML use cases, moving through data access and preparation, building and validating predictive models, and finally deploying and administering models and workflows at an enterprise level. The internship was conducted entirely in a virtual, simulation/lab-based environment provided by EduSkills Academy and RapidMiner, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Data Science Platform: RapidMiner Studio and RapidMiner AI Hub (Altair)
- Data Engineering: Data access, pivot tables, multi-dataset handling, macros, web APIs, scripting
- Machine Learning: Classification, regression, clustering, association rules, cross-validation, feature engineering, time-series forecasting
- Advanced Data Preparation: Loops and branches, regular expressions, advanced text processing, exception handling, logging
- Model Deployment: Web application deployment and model monitoring
- Platform Administration: RapidMiner Studio installation, RapidMiner AI Hub administration, RapidMiner Radoop, real-time scoring, RapidMiner Marketplace

4.7 System Architecture

The general workflow followed during the internship for a data science task on the RapidMiner platform can be summarized in the following layers:

- Business Understanding Layer: Identifying a business problem suitable for an AI/ML solution.
- Data Layer: Accessing, loading, cleaning, and transforming raw data from multiple sources.
- Modelling Layer: Building and validating machine learning models (classification, regression, clustering) within RapidMiner Studio.
- Optimization Layer: Tuning models through cross-validation, parameter optimization, and feature engineering.
- Deployment Layer: Publishing models and workflows as scored applications and monitoring them for performance.
- Administration Layer: Managing the underlying RapidMiner AI Hub infrastructure, including Radoop integration and marketplace extensions.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, weekly assessments, and project submissions, as summarized in the table below.

Week	Module	Description
1	Applications & Use Cases Professional	Learning to identify business problems suitable for AI and machine learning.
2	Data Engineering Professional	Learning to access, load, transform, and calculate data.
3	Machine Learning Professional	Building foundational machine learning skills.
4	Data Preparation Master	Mastering advanced data handling and automation.
5	Machine Learning Master	Learning advanced machine learning techniques.
6	Applications & Use Cases Master	Deploying and monitoring ML models.
7	Platform Administration Master	Configuring and managing RapidMiner AI Hub.
8	Platform Administration Master	Enterprise-level platform operations.

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and assignments. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across weekly assessments, project submissions, and the final test. There was no stipend associated with this internship.

4.9 Expected Outcomes

- Ability to independently identify business problems suitable for AI/ML solutions.
- Practical understanding of data access, transformation, and multi-dataset handling using an industry data science platform.
- Hands-on exposure to building, validating, and optimizing classification, regression, and clustering models.
- Capability to apply advanced data preparation techniques such as loops, branches, and regular expressions.

- Understanding of model deployment and monitoring in a web application setting.
- Skill to administer an enterprise data science platform, including RapidMiner AI Hub, Radoop, and real-time scoring.
- A completed, assessed internship demonstrating end-to-end data science competency, graded 'O - Outstanding' by EduSkills Academy.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Certificate of Virtual Internship issued by EduSkills Academy and Siemens Industry Software, confirming successful completion of the 8-week AICTE - EduSkills Virtual Internship Program in Data Science Master (Certificate ID: 44d7b3aef729cae3c5be, Grade: O), are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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7. RapidMiner Documentation, <https://docs.rapidminer.com/>
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